

Unload the Dryer SOP (1x Unload, in situ)

One episode empties one finished dryer where it is installed: six garments come out of the drum and go into the hamper, each shaken once on the way, the lint filter is cleared, the door is closed, and the hamper is carried to the table. The base is **passive**: it has no drive of its own, so it is pushed by hand up to the dryer and locked there. The table the hamper is carried to stands at the dryer, within reach of the arms, so nothing is carried away from the machine.

The dryer is worked **as found**. The cycle has finished and the machine is off. The door already stands open to the left when recording starts, the six garments lie loose on the drum floor as the cycle left them, and the lint filter sits in its slot with a sheet of lint on it. The hamper stands empty on its stand to the right of the drum mouth, and the table in front of the dryer is empty apart from the lint bin.

This is an in-situ task, and three things follow from that. First, the drum is deep and the arms cannot reach its back, so **no gripper reaches into the drum for a garment. A garment is dragged to the mouth, then lifted out.** Second, **the dryer, the stand, and the table are never leaned on and never pushed**: no gripper, wrist, or forearm rests on the door, the mouth rim, the stand, or the table, and the only push in this SOP is the door being closed. Third, **nothing is thrown**: every garment is held over the hamper and let go, and the hamper is lifted, carried level, and set down.

The order never changes: **all six garments, then the lint filter, then the door, then the hamper.** The filter is cleared with the door open because the filter sits in the door opening. The door is closed before the hamper moves, so the hamper is never carried while the door stands out.

Arm roles are fixed and never swap. **The right gripper does all the garment work, the lint peel, the door press, and the hamper carry. The left gripper is the helper at the mouth**: it frees a second garment that comes out tangled, it pulls the lint filter out and holds it for the peel and puts it back, and it pushes the door round. Neither gripper does the other's work.

Setup

Complete the base positioning and all the checklists before starting an episode.

Cell configuration

- **Environment camera**: 900 mm.
- **cell_type**: bimanual

Base positioning

The base is **passive**. It has no drive of its own, it is pushed into place by hand, and it is never steered, nudged, or repositioned once recording starts. It is parked once, before recording, and does not move again until the episode is over.

1. Push the base by hand up to the near edge of the table and stop it **square to the front of the dryer**, so the front of the machine runs straight across the frame and neither end of the table sits nearer than the other.
2. Stop it **centered on the drum mouth**, so the left rim and the right rim of the mouth are the same distance out from the middle of the base.
3. Stop it **close enough** that the **right gripper** reaches through the mouth and down onto the front of the drum floor without the arm extending.
4. Check the **open door**: standing open to the left, wider than a right angle, it is clear of the left arm, the left side of the base, and the left arm's path to the mouth.
5. Check the **filter slot**: the **left gripper** comes down onto the filter tab at the bottom of the mouth from above, pulls the filter straight up out of its slot, and holds it upright in front of the mouth, all without extending and without touching the mouth rim.
6. Check the **hamper**: the **right gripper** reaches over the whole open top of the hamper on its stand, and closes on the carry handle at its top middle, without extending.
7. Check the **table**: the **right gripper** reaches the lint bin from above and lowers the hamper onto the hamper spot, without extending and without touching the closed door or the table edge.
8. Check the **door**: the **left gripper** reaches the door's inner face near the hinge, and the **right gripper** reaches the door's front face at its free edge once the door is round in front of the mouth, without extending.
9. Lock or brake the base. Push it firmly once by hand: it must not roll, creep, or turn.
10. If any of lines 1 to 8 fails, push the base to a new park by hand and start again at line 1. Do not work a station the arms cannot reach comfortably.

The base stays locked and still for the whole episode. Nothing moves it: no arm leans on the dryer, the stand, or the table hard enough to shift it, nothing and nobody touches it, and it is never repositioned mid-task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the drum mouth and its frame includes the open door, the whole mouth, the front of the drum floor, the filter slot, the hamper on its stand, and the whole table with the lint bin.
3. The camera reads the drum floor through the mouth, so a garment still lying in the drum is visible.

4. The camera reads the lint filter held upright in front of the mouth, so the lint face is visible and a peeled mesh can be told from one still carrying lint.
5. The camera reads the open top of the hamper from above, so a garment lying inside and a garment hanging over the rim can be told apart.
6. The camera reads the door front once it is shut, so a gap along the door edge is visible.
7. Both arms are at home with grippers open.
8. The **right arm** reaches the front of the drum floor through the mouth, the whole open top of the hamper, the hamper carry handle, the lint bin, the hamper spot, and the door's front face at its free edge, without extending to a joint limit. It reaches neither the filter tab nor the door's inner face.
9. The **left arm** reaches the mouth, the filter tab, and the door's inner face near the hinge without extending to a joint limit. It reaches nothing on the stand or the table.
10. The two arms do not collide, and the open door is clear of both arms' paths to the mouth and to the hamper.
11. If a place cannot be reached, re-park the base by the Base positioning steps until lines 8 to 10 hold.

Materials checklist

1. The **dryer** is a front-loading dryer standing where it is installed, on its pedestal or as the top machine of a stack, so the bottom of the drum mouth sits about level with the arms. It is not moved, not leaned on, and not pushed at any point. The cycle has finished and the machine is off.
2. The **door** is hinged on the **left** of the machine. At the start of the episode it stands open to the left, wider than a right angle, and it stays there on its own. It shuts with a latch that holds it closed, and a correctly shut door sits flush with the front of the machine all round.
3. The **drum** is still. Six **garments** lie loose on the **drum floor**: three cotton t-shirts and three hand towels, all dry, none rolled, knotted, or tucked into another. Every garment has some part lying in the front half of the drum floor. Nothing hangs out of the mouth.
4. The **lint filter** sits fully seated in its **filter slot** at the bottom of the door opening, with its **tab** standing up where the left gripper can pinch it. It lifts straight up out of the slot and lowers straight back in without catching. A **lint sheet** covers most of its mesh on the lint face.
5. The **hamper** is a small, light, open-top laundry basket with one **carry handle**: a stiff hoop fixed across the top from its near rim to its far rim, standing up. It stands empty on the **hamper stand**, handle up, and it does not rock or slide when a garment drops into it.
6. The **hamper stand** stands on the floor to the right of the drum mouth, in front of the right part of the machine, beyond where the door's free edge reaches when the door swings shut. Its top is level with the table top. It does not move when the hamper is lifted off it.
7. The **table** stands on the floor in front of the dryer, directly below the door opening, between the dryer and the base. Its top is lower than the bottom edge of the open door, so the door

swings shut clear over it. Its right end stops short of the hamper stand. It does not move when the hamper is set down on it.

8. The **lint bin** stands open at the right end of the table, empty. It is heavy enough not to tip when lint drops into it.
9. The **hamper spot** is the rest of the table top, to the left of the lint bin, and it is clear.
10. The floor around the stand, the table, and the base is clear and dry.

Workspace layout

- **Dryer:** straight ahead of the parked base, on its pedestal or stack, drum mouth about level with the arms.
 - **Mouth:** the round door opening. The **bottom rim** of the mouth is where a garment is lifted over on its way out. The **filter slot** runs along the bottom of the opening with the filter tab standing up in it.
 - **Drum floor:** the bottom of the drum, where the six garments lie. Only its front is within reach.
 - **Door:** hinged on the left, standing open to the left wider than a right angle. Its **inner face** faces the mouth. Its **front face** faces out. Its **free edge** is the edge away from the hinge, where the latch is.
- **Table:** low, in front of the dryer under the door opening, between the machine and the base.
 - **Lint bin:** at the right end of the table.
 - **Hamper spot:** the rest of the table top, left of the lint bin.
- **Hamper stand:** to the right of the table, in front of the right part of the machine, same height as the table.
 - **Hamper:** on the stand, carry handle up. The handle splits its open top into a **left half**, nearer the mouth, and a **right half**.

The right arm works the mouth, the hamper, the lint bin, the hamper spot, and the door's free edge. The left arm works the mouth, the filter slot, and the door's inner face, and stays on its own side of the machine.

Arm assignments

- **The right gripper owns the garments, the lint, the door press, and the hamper.** It drags every garment to the mouth, lifts it out, shakes it once over the hamper, and lets it go. It peels the lint off the filter and drops it in the lint bin. It presses the door shut at its free edge. It lifts the hamper by its carry handle and carries it to the hamper spot. It never touches the filter tab, the door's inner face, or the hamper rim.
- **The left gripper is the helper at the mouth.** It frees a second garment that comes out with the one in the right gripper and drops it back into the drum. It pulls the lint filter out, holds it

for the peel, and seats it back. It pushes the door round. It never lifts a garment out of the drum, never touches the hamper, the lint bin, or the table, and never presses the door shut.

- A garment is released only over the left half of the hamper. Lint is released only over the lint bin. The hamper is released only when it is resting on the hamper spot. Never open a gripper over open space, over the drum, over the stand, or over the floor.

Vocabulary

- **In situ:** the dryer is worked where it is installed. Nothing is taken away from the machine, and nothing about the laundry area is rearranged for the task.
- **Passive base:** the base has no drive. It is pushed by hand to its park before recording, locked there, and it does not move at all during the episode.
- **Drum:** the round tumbling chamber inside the dryer. The **drum floor** is its bottom, where the garments lie after the cycle.
- **Mouth:** the round door opening at the front of the drum. The **bottom rim** is the lower edge of the opening.
- **Garment:** one of the six pieces of laundry, a t-shirt or a hand towel.
- **Nearest part:** the part of a garment that lies closest to the mouth, whatever it is: a hem, a sleeve, a cuff, or a corner. The right gripper pinches the nearest part and nothing else.
- **Drag to the mouth:** the right gripper pinches the nearest part and pulls the garment forward along the drum floor until the pinched part is at the bottom rim. The garment is not lifted while it is dragged. **Unvalidated.**
- **Lift out:** with the pinched part at the bottom rim, the right gripper rises straight up and forward until the whole garment hangs from it clear of the bottom rim and the filter tab.
- **Tangled:** a second garment comes out of the mouth with the one in the right gripper, caught in it or lying over it.
- **One shake:** with the garment hanging over the left half of the hamper, one quick drop and rise of the right gripper, about a hand's length down and straight back up. Exactly one. Not two, not a sideways flick, and not over the drum or the table.
- **Inside the hamper:** the whole garment lies below the hamper rim. No part hangs over the rim or over the carry handle.
- **Lint filter:** the flat mesh screen that sits in the **filter slot** at the bottom of the door opening. Its **tab** is the grip at its top that stands up out of the slot. Its **lint face** is the face of the mesh that carries the sheet of lint. **Unvalidated:** which face carries the lint.
- **Lint sheet:** the layer of lint lying on the lint face. It comes off the mesh in a sheet when peeled from one corner.
- **Peel:** the right gripper pinches the lint sheet at its top corner and draws it down and off the mesh in one pull, without rubbing the mesh.
- **Clear mesh:** the mesh shows through over its whole face, with no patch of lint left on it.

- **Seated:** the lint filter is fully down in its slot, its tab stands where it stood at the start, and nothing of the filter stands proud of the slot.
- **Door inner face:** the side of the door that faces the mouth when the door stands open. The left gripper pushes here, near the hinge.
- **Door front face:** the outer side of the door. The right gripper presses here, at the **free edge**, the edge away from the hinge where the latch is.
- **Round:** the door has swung from standing open to hanging in front of the mouth, touching or nearly touching the front of the machine, but not yet latched.
- **Shut:** the door is latched, sits flush with the front of the machine all round, no gap shows along its edge, and it does not spring back when the right gripper comes off.
- **Hamper:** the small open-top laundry basket the garments go into.
- **Carry handle:** the stiff hoop fixed across the top of the hamper from its near rim to its far rim. The right gripper closes on it at its top middle. **Unvalidated:** grip on the hoop and the weight of the loaded hamper.
- **Left half / right half:** the two halves of the hamper's open top, either side of the carry handle. The left half is the half nearer the mouth. Every garment goes into the left half.
- **Hamper stand:** the stand to the right of the table where the hamper starts.
- **Hamper spot:** the part of the table top left of the lint bin where the hamper is set down.
- **Lint bin:** the open bin at the right end of the table where the lint goes.
- **Resting:** the thing has stopped moving on the surface below it and stays still for 2 seconds after the gripper opens. A resting hamper stands flat on all of its base and does not rock.
- **Clear of:** not touching. A gap shows between the two things.

Handling standard

- Move exactly one garment at a time.
- No gripper reaches into the drum for a garment. Take the nearest part, drag to the mouth, then lift out.
- Every garment is held over the left half of the hamper, shaken once, and let go there. Nothing is thrown, and nothing is shaken anywhere else.
- Lint goes only in the lint bin.
- The filter comes straight up out of its slot and goes straight back down into it. It is never forced.
- The door is touched only in Step 3, once the filter is seated and both grippers are out of the mouth.
- The hamper is touched only in Step 4, once the door is shut. It is lifted by its carry handle, carried level and low, and lowered until it rests. It is never dragged across the stand or the table.

- Nothing rests on the dryer, the stand, or the table. No gripper, wrist, or forearm leans on the door, the mouth rim, the stand, or the table, and no push is ever hard enough to move any of them.

Steps

Step 1: Take the six garments out of the drum into the hamper

Goal: the drum is empty and all six garments lie inside the hamper, each shaken once on the way.

Work one garment at a time. Take the garment with the **nearest part** first. Never dig under one garment for another.

1.1 Drag the garment to the mouth

- The **right gripper** enters the mouth level and pinches the nearest part of the garment.
- **Drag** the garment forward along the drum floor until the pinched part is at the bottom rim. Keep the garment on the drum floor while it moves.
- If the nearest part slips out of the gripper, pinch the part that is now nearest and drag again.

1.2 Lift the garment out

- **Lift out:** the **right gripper** rises straight up and forward until the whole garment hangs from it, clear of the bottom rim and the filter tab.
- Carry it to the right, hanging, until it is over the **left half** of the hamper, with the gripper a little above the rim so the garment hangs down into the hamper.
- If a second garment is **tangled** with it, stop with the garment hanging just outside the mouth. The **left gripper** pinches the second garment where it hangs, pulls it free with a short pull, holds it over the mouth, and lets it drop back onto the drum floor. Then the **right gripper** carries on to the hamper. The right gripper never shakes a tangle apart.

1.3 Shake it once and let it go

- Over the left half of the hamper, give the garment **one shake**: one quick drop and rise of the **right gripper**, about a hand's length, straight down and straight back up.
- Open the **right gripper** and let the garment fall into the left half. Do not throw it and do not push it down.
- Lift the **right gripper** straight up and clear of the hamper.

Check: the garment lies **inside the hamper**. If any part hangs over the rim or over the carry handle, the **right gripper** pinches that part, lifts it up over the rim, and lets it fall inside. A garment that has landed on the stand or the table is picked up by any part with the **right gripper**, held over the left half, and let go again without a second shake. A garment that has fallen to the floor is left where it is.

Go back to 1.1 for the next garment until no garment is left in the drum.

Check: look into the drum. No garment lies on the drum floor, and none is caught on the bottom rim or in the mouth. If one is left, go back to 1.1 for it.

Expected state: the drum is empty, six garments lie inside the hamper on its stand, the door still stands open to the left, and both grippers are empty and out of the mouth.

Step 2: Clear the lint filter

Goal: the lint is off the filter and in the lint bin, and the filter is seated back in its slot.

2.1 Pull the filter out

- The **left gripper** comes down onto the filter **tab** from above and pinches it.
- Pull the filter **straight up** out of the slot, in one movement, until it is fully clear of the slot. If it catches, lower it back and pull straight up again. Never pull at an angle and never force it.
- Hold the filter upright in front of the middle of the mouth, and turn it a quarter turn with the wrist so its **lint face** points to the right, toward the right gripper.
- The **left gripper** holds the filter still there until 2.3.

Check: the whole filter stands clear of the slot, upright, still, with the lint face to the right.

2.2 Peel the lint off and bin it

- The **right gripper** comes in from the right and pinches the **lint sheet** at its top corner, next to the tab.
- **Peel** the sheet down and off the mesh in one pull. Do not rub the mesh and do not scrape it with the gripper.
- Carry the lint over the **lint bin** and open the gripper so it drops in. Lift the gripper straight up.
- If lint is still on the mesh, pinch the biggest patch that is left and peel again. Repeat until the mesh is clear.

Check: the mesh is **clear mesh** on its whole face, and all the lint is in the lint bin. If a patch is left, the **right gripper** peels it. If lint has dropped on the table or the stand, the **right gripper** picks it up and drops it in the bin.

2.3 Seat the filter back

- The **left gripper** turns the filter a quarter turn back so it faces as it did in the slot, and lowers it **straight down** into the slot, tab up, until it stops.
- If it stops short, lift it straight up, line it up over the slot again, and lower it again. Never push it down at an angle and never force it.
- Open the **left gripper** and lift it straight up, clear of the mouth.

Check: the filter is **seated**: fully down, tab standing where it stood at the start, nothing proud of the slot.

Expected state: the filter is seated with a clear mesh, the lint is in the lint bin, the door still stands open to the left, and both grippers are empty and out of the mouth.

Step 3: Close the door

Goal: the door is **shut** and flush with the front of the machine.

The door swings from the left, so the **left gripper** pushes it round. It latches at its free edge on the right, so the **right gripper** presses it home.

3.1 Push the door round

- Check first that both grippers are empty and out of the mouth, and that the filter is seated.
- With the **left gripper** closed, set it on the door's **inner face** near the hinge.
- Push steadily, so the door swings **round** toward the mouth in one smooth move. Do not push fast and do not let the door swing on its own.
- Stop when the door hangs in front of the mouth, touching or nearly touching the front of the machine.
- Draw the **left gripper** straight back off the door and keep it clear until the episode ends.

Check: the door hangs in front of the mouth and nothing is caught between the door and the machine. If a garment part or the filter tab is caught, the **left gripper** pushes the door back open, the **right gripper** clears the garment into the hamper or the **left gripper** reseats the filter, and the door is pushed round again.

3.2 Press the door shut

- With the **right gripper** closed, press the door's **front face** flat at the **free edge**, straight back toward the machine.
- Press until the latch holds. Hold the press for 2 seconds.
- Press straight back only. Do not slam it, do not press at an angle, do not press on the door glass, and do not press hard enough to move the machine.
- Draw the **right gripper** straight back off the door.

Check: the door is **shut**: flush with the front all round, no gap along its edge, and it does not spring back when the **right gripper** comes off. If it springs back or a gap shows, the **right gripper** presses it home once more at the free edge.

Expected state: the door is shut, the drum is empty behind it, the six garments are in the hamper on its stand, and both grippers are clear of the door.

Step 4: Carry the hamper to the table

Goal: the hamper stands resting on the hamper spot with all six garments inside.

- The **right gripper** comes down from above and closes on the **carry handle** at its top middle.

- Lift the hamper **straight up** until its base is just clear of the stand, and no higher.
- Carry it **level and low** to the left, over the table, until it hangs over the **hamper spot**, clear of the lint bin. Do not swing it, do not tip it, and do not drag it across the stand or the table.
- Lower it **straight down** until it is **resting** on the hamper spot, then open the gripper and lift it straight up.
- The **left gripper** does not touch the hamper at any point.

Check: the hamper stands flat on the hamper spot, clear of the lint bin and the table edge, with all six garments inside. If it rocks or sits on the bin, the **right gripper** closes on the handle, lifts it just clear, and lowers it again. If a garment has fallen out onto the table, the **right gripper** picks it up by any part, holds it over the left half of the hamper, and lets it go.

Expected state: the hamper is resting on the hamper spot with six garments inside, the stand is empty, the door is shut, and the lint is in the lint bin.

Step 5: End the episode

Goal: the unload is complete, the dryer is shut, and both arms are safely home.

- Confirm the door is shut and flush, and the filter tab stands seated at the bottom of the door opening behind it.
- Confirm the hamper rests on the hamper spot with all six garments inside and the hamper stand is empty.
- Confirm the lint is in the lint bin and none lies on the table, the stand, or the floor.
- Confirm the base has not moved: it is still square to the machine and still locked.
- Return both arms home, then stop recording.

After the episode: reset the workspace

This reset is not recorded. The base stays parked and locked through the reset, and is only pushed away by hand once the station is set for the next episode.

1. Open the door to the left, wider than a right angle, and check it stays there on its own.
2. Lift the hamper off the hamper spot and take the six garments out. Shake each one open and put all six back loose on the drum floor, spread so that every garment has some part lying in the front half. None rolled, knotted, or tucked into another, and nothing hanging out of the mouth.
3. Stand the hamper empty on the hamper stand, carry handle up.
4. Empty the lint bin and put it back at the right end of the table. Check the hamper spot is clear.
5. Pull the lint filter out, press a fresh lint sheet onto its lint face so it covers most of the mesh, and seat the filter back in its slot, tab up.

6. Replace any garment that is torn, damp, or shedding, and any lint sheet that will not stay on the mesh.
7. Check the filter still lifts straight out and lowers straight back without catching, the door still stands open on its own and latches when pressed, and the stand and the table have not shifted.
8. Check the base is still square, centered, and locked, and push it firmly once by hand to confirm it does not roll. If it has moved, park it again by the Base positioning steps.
9. Run the Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Base moved during the episode

- **Visible cue:** the base rolls, creeps, or turns after recording starts, the mouth shifts in the frame, or an arm pushes off the dryer, the stand, or the table hard enough to move the base.
- **SOP rule broken:** Step 5 and the Base positioning rule it confirms, the base is parked and locked before recording and does not move at all during the episode.
- **Coaching note:** park it, lock it, push-test it. If the park is wrong, fix it before recording, never during.

Violation: Dryer, stand, or table leaned on, pushed, or moved

- **Visible cue:** a gripper, wrist, or forearm rests on the mouth rim, the door, the stand, or the table; an arm pushes off any of them; or the stand, the table, or the machine visibly shifts under a push.
- **SOP rule broken:** Steps 1 to 4, nothing rests on the dryer, the stand, or the table, and the only push is the door being closed in Step 3.
- **Coaching note:** nothing at the station holds an arm up. Take the weight on the arm.

Violation: Work order broken

- **Visible cue:** the work does not run all six garments, then the lint filter, then the door, then the hamper. For example the filter comes out while a garment is still in the drum, the door is pushed before the filter is seated, or the hamper is lifted while the door still stands open.
- **SOP rule broken:** Steps 1 to 4, the order is garments, filter, door, hamper.
- **Coaching note:** say the next stage out loud before you reach for anything. The order never changes.

Violation: Garment dug out or not taken by its nearest part

- **Visible cue:** the right gripper pinches a garment that is not the one with the nearest part, pushes one garment aside to get at another, or pinches a garment somewhere other than its nearest part.
- **SOP rule broken:** Step 1.1, take the garment with the nearest part first, pinch it by that part, and never dig under one garment for another.
- **Coaching note:** the nearest bit of cloth is the one you take. The rest of the pile is not yours yet.

Violation: Garment reached for or lifted instead of dragged

- **Visible cue:** the right gripper reaches deep into the drum for a garment, lifts a garment straight up from where it lies on the drum floor instead of dragging it to the bottom rim first, or drags it out over the filter tab without lifting it clear.
- **SOP rule broken:** Steps 1.1 and 1.2, drag the garment along the drum floor to the bottom rim, then lift out straight up and forward, clear of the rim and the filter tab.
- **Coaching note:** pull it to you, then lift. The arm never goes into the drum after cloth.

Violation: Tangled garment shaken or pulled apart by the right gripper

- **Visible cue:** a second garment comes out with the one in the right gripper and the right gripper shakes, tugs, or swings it off instead of stopping, or the second garment is carried to the hamper still caught on the first.
- **SOP rule broken:** Step 1.2, stop with the garment just outside the mouth and let the left gripper pull the second garment free and drop it back onto the drum floor.
- **Coaching note:** stop, and let the left hand take the extra one. The right hand never shakes over the drum.

Violation: Second garment dropped outside the drum

- **Visible cue:** the left gripper frees a tangled garment and lets it go outside the mouth, so it lands on the table, the stand, or the floor instead of back on the drum floor.
- **SOP rule broken:** Step 1.2, the left gripper holds the freed garment over the mouth and lets it drop back onto the drum floor.

- **Coaching note:** hold it over the mouth before you open. It goes back in the drum, nowhere else.

Violation: Garment not shaken once over the hamper

- **Visible cue:** a garment is let go with no shake, is shaken two or more times, is flicked sideways, or is shaken over the drum, the table, or the floor instead of over the left half of the hamper.
- **SOP rule broken:** Step 1.3, over the left half of the hamper, give the garment one shake, one quick drop and rise of the right gripper, then let it go.
- **Coaching note:** one down, one up, then open. Count it.

Violation: Garment thrown or released outside the left half

- **Visible cue:** the right gripper opens before the garment is over the left half of the hamper, flings the garment in, lets it go over the right half, or pushes it down into the hamper after release.
- **SOP rule broken:** Step 1.3, open the right gripper over the left half of the hamper and let the garment fall. Do not throw it and do not push it down.
- **Coaching note:** over the left half, still, then open. Gravity does the rest.

Violation: Drum not confirmed empty

- **Visible cue:** the right gripper leaves the drum for the filter while a garment still lies on the drum floor or hangs on the bottom rim, or fewer than six garments are in the hamper when the filter comes out.
- **SOP rule broken:** Step 1, look into the drum and confirm no garment is left before moving to Step 2.
- **Coaching note:** look in the drum before you touch the filter. Six in the hamper, none in the drum.

Violation: Filter pulled wrongly

- **Visible cue:** the right gripper takes the filter tab, the filter is pulled at an angle, pulled part way and let go, forced when it catches, or lifted with the gripper pinching the mesh instead of the tab.
- **SOP rule broken:** Step 2.1, the left gripper pinches the tab from above and pulls the filter straight up in one movement, and if it catches, lowers it and pulls straight up again.
- **Coaching note:** tab only, straight up. If it sticks, back down and straight up again.

Violation: Filter not held upright and still for the peel

- **Visible cue:** the left gripper does not turn the filter so the lint face points right, lets it lean or wave during the peel, sets it down on the table or the mouth rim, or lets go of it before it is back in the slot.

- **SOP rule broken:** Steps 2.1 and 2.2, the left gripper holds the filter upright and still in front of the mouth, lint face to the right, until the mesh is clear.
- **Coaching note:** the left hand is the stand for the filter. It does not move until the mesh is clear.

Violation: Lint not peeled in a sheet or mesh left with lint

- **Visible cue:** the right gripper rubs, scrapes, or picks at the mesh instead of pinching the top corner and peeling down, the left gripper does the peel, or the filter goes back into its slot with a patch of lint still on it.
- **SOP rule broken:** Step 2.2, the right gripper pinches the lint sheet at its top corner and peels it down and off in one pull, and repeats on any patch left until the mesh is clear.
- **Coaching note:** corner, then one long peel. Look at the mesh before the filter goes home.

Violation: Lint dropped anywhere but the lint bin

- **Visible cue:** the right gripper opens on the lint over the hamper, the drum, the table, the stand, or the floor, or lint falls from the gripper on the way and is not picked up and binned.
- **SOP rule broken:** Step 2.2, carry the lint over the lint bin and open the gripper there.
- **Coaching note:** lint goes in the bin, every piece. Clean laundry does not get lint back.

Violation: Filter not seated

- **Visible cue:** the filter is lowered at an angle, forced down when it stops short, left standing proud of the slot, left out on the table, or the door is pushed with the filter not fully down.
- **SOP rule broken:** Step 2.3, lower the filter straight down into the slot, tab up, until it stops, and confirm it is seated before the door is touched.
- **Coaching note:** straight down until it stops, then look at the tab. A proud filter jams the door.

Violation: Door touched early or pushed with the mouth blocked

- **Visible cue:** a gripper takes hold of the door or pushes it during Steps 1 or 2, or the left gripper pushes the door round while a gripper is still in the mouth, a garment is still hanging, or the filter is not seated.
- **SOP rule broken:** Step 3.1, the door is touched only once the filter is seated and both grippers are empty and out of the mouth.
- **Coaching note:** filter in, hands out, then the door.

Violation: Door pushed round wrongly

- **Visible cue:** the right gripper pushes the door round, the door is pushed fast, let swing on its own, pushed at the free edge instead of near the hinge, or hits the machine.
- **SOP rule broken:** Step 3.1, the left gripper on the inner face near the hinge pushes the door round steadily in one smooth move and stops it in front of the mouth.

- **Coaching note:** near the hinge, slow and steady. A fast door bounces open.

Violation: Door not pressed shut

- **Visible cue:** the episode goes on with the door open or ajar, a gap shows along its edge, the latch has not held, the door springs back when the right gripper comes off, or the press is made at an angle, on the door glass, or by the left gripper.
- **SOP rule broken:** Step 3.2, the right gripper presses the door's front face flat and straight back at the free edge, holds the press for 2 seconds, and the door sits flush with the latch holding.
- **Coaching note:** flat face, straight back, hold two seconds, then watch it before you move away.

Violation: Hamper touched or moved before the carry

- **Visible cue:** a gripper holds, steadies, or nudges the hamper during Steps 1 to 3, the hamper shifts on its stand under a garment drop or a knock, or the left gripper touches the hamper at any time.
- **SOP rule broken:** Steps 1 to 3, the hamper is touched only in Step 4, and only by the right gripper.
- **Coaching note:** the hamper is a target until Step 4. Nothing holds it.

Violation: Hamper carried wrongly

- **Visible cue:** the hamper is lifted by its rim instead of the carry handle, lifted with both grippers, lifted high, tipped, swung, or dragged across the stand or the table on the way.
- **SOP rule broken:** Step 4, the right gripper closes on the carry handle at its top middle, lifts the hamper just clear of the stand, and carries it level and low to the hamper spot.
- **Coaching note:** handle in the middle, just off the stand, level all the way.

Violation: Hamper not set down resting on the hamper spot

- **Visible cue:** the right gripper opens before the hamper is resting, the hamper lands on the lint bin, half off the table, or rocking, or it is pushed across the table to line it up.
- **SOP rule broken:** Step 4, lower the hamper straight down onto the hamper spot until it is resting, then release, and fix a bad set-down by lifting it clear and lowering it again.
- **Coaching note:** all the way down, then open. Lift to fix, never slide.

Violation: Wrong arm used

- **Visible cue:** the left gripper lifts a garment out of the drum, peels lint, presses the door shut, or touches the hamper, the lint bin, or the table; or the right gripper takes the filter tab, pushes the door's inner face, or frees a tangled garment.
- **SOP rule broken:** Steps 1 to 4, the right gripper does the garments, the peel, the door press, and the hamper carry; the left gripper frees a tangle, works the filter, and pushes the

door round.

- **Coaching note:** right hand carries and presses. Left hand helps at the mouth.

Violation: Garment, lint, or hamper dropped or knocked

- **Visible cue:** a garment falls from the right gripper onto the stand, the table, or the floor; the filter is dropped; the lint bin is knocked over; or the hamper is dropped or a garment falls out of it during the carry.
- **SOP rule broken:** Steps 1 to 4, hold every garment until it is over the left half, hold the filter until it is seated, and carry the hamper level and low until it is resting.
- **Coaching note:** carry low and level, and open only over the place it goes.

Violation: Correction made without confirming the check

- **Visible cue:** a step's check is skipped: the episode moves on with a garment hanging over the hamper rim or the handle, lint still on the mesh, the filter proud of its slot, the door ajar, or the hamper rocking, and nothing is fixed.
- **SOP rule broken:** Steps 1.3, 2.2, 2.3, 3.2, and 4, each check is confirmed before the next stage begins, and a bad result is fixed the way the step says.
- **Coaching note:** look at it before you reach for the next one. Fix it now, not at the end.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a garment still in the drum, the door open, the hamper still on its stand, lint on the table, or an arm away from home.
- **SOP rule broken:** Step 5 (confirm the door shut and the filter seated, the hamper resting on the hamper spot with six garments inside, the lint in the bin, and the locked base; return both arms home; then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Base fault:** a brake that releases on its own, or a caster that seizes so the base cannot be parked square.
- **Defective object:** a torn or damp garment, a lint sheet that falls off the mesh on its own, a filter that jams in its slot under a straight pull, a hamper handle that comes loose, or a lint bin that tips under its own weight. Replace or refix it before the next episode.

- **Machine or station fault:** a door that will not stay open on its own or will not latch under a flat press, a drum that turns on its own, or a stand or table that slides under a correct level set-down.

Annotation subtasks (from SOP)

1. Drag one garment to the mouth and lift it out
2. Free a tangled second garment with the left gripper and drop it back into the drum
3. Shake one garment once over the hamper and let it go
4. Pull the lint filter out of its slot and hold it upright
5. Peel the lint off the filter and drop it in the lint bin
6. Seat the lint filter back in its slot
7. Push the door round with the left gripper
8. Press the door shut with the right gripper
9. Carry the hamper to the hamper spot
10. Return both arms home and end the episode